

# Diagnostic Assessment of AI-Generation Risk and Journal Readiness for the EXOhSPEC Manuscript

## AI-generation signature assessment

My overall reading is **not** “this looks fabricated” or “this looks machine-written end to end”. The technical content, instrument-specific detail, equations, calibration logic, run durations, and single-run caveats all read like domain work by real researchers. The main risk is narrower: **substantial portions of the framing prose read strongly AI-polished**. In particular, the manuscript repeatedly uses contrastive scaffolding, self-conscious claim-scoping, and polished “meta-writing” that explains what the paper *is not* doing almost as often as it reports what it *did* do. That is exactly the kind of rhetorical smoothness that makes editors and reviewers suspect LLM assistance even when the underlying science is real. The **highest-risk sections are the abstract, the later Introduction, much of the Discussion, the Planned validation section, and the Conclusions**. By contrast, the core methods/results sections are materially safer because they are anchored in equations, hardware, thresholds, command counts, and comparative runs. [filecite turn0file0](#) [filecite turn0file1](#)

The strongest recurring marker is **contrastive “not X, rather Y” framing**. It appears throughout the manuscript and, in cumulative effect, feels algorithmically overused. Representative examples include: **“the first directly measured, rather than inferred, quantification of the closed loop’s benefit in this programme”** (Abstract, p. 1); **“The relevant instrumentation literature can be read as a sequence of methodological lessons rather than as performance benchmarks”** (Introduction, pp. 1–2); **“image position, optical path, actuator commands and environmental context must be retained together rather than in isolation”** (Introduction, p. 2); **“applied directly across configurations rather than requiring a cross-run percentage correction”** (Section 2.2, p. 3); **“This is a useful negative result rather than a null one”** (Section 4.3, p. 7); **“direct measurement rather than by design intent alone”** (Discussion, p. 7); and **“measured directly on the same hardware rather than asserted from the architecture alone”** (Discussion, p. 9). One or two such contrasts are normal; this manuscript uses them so frequently that they become a stylistic signature. [filecite turn0file0](#) [filecite turn0file1](#)

A second strong marker is the repeated use of **paragraph-final “claim restatement” sentences**. These do not add new evidence; they summarise the paragraph’s own point in a polished, self-validating voice. Examples include the opening paragraph’s closing sentence, **“our data measure image-centroid displacement and one monitored interferometric path, not a complete stellar radial-velocity error budget”** (Introduction, p. 1); the paragraph ending **“These channels are recorded synchronously so that image position, optical path, actuator commands and environmental context are always interpreted together rather than as single-variable causal explanations of image motion”** (Section 2.1, p. 3); the Section 4.2 close, which reiterates that image control and optical-path evolution are distinct outcomes (p. 7); and the Discussion close that broadens the approach to **“other compact or cost-constrained spectrographs”** (p. 9). This kind of paragraph architecture is common in LLM-polished prose because it produces a smooth didactic cadence, but it often feels over-managed in a specialist instrumentation paper. [filecite turn0file0](#)

A third marker is the paper's reliance on **abstract nominalisations and branded system adjectives**. Phrases such as "**guarded architecture**", "**guarded, event-driven controller**", "**guarded, self-identifying, dual-bandwidth controller**", "**self-validating philosophy**", "**evidence-led questions**", "**decision layer**", "**reduced-authority mode**", and "**fully passive mode**" are not individually wrong, but together they create a "designed register" rather than a natural technical voice. The repeated self-description makes the manuscript sound as though it is presenting a product or framework. This is especially noticeable in the abstract's final sentence and in the Discussion paragraph beginning "**The scientific value of this work to the wider instrumentation community is that it demonstrates...**" (p. 9). [filecite turn0file0](#)

A fourth marker is the manuscript's fondness for **triplet and layered-list constructions**. For example: "**an observable... one or more actuators... and a decision layer**" (Introduction, p. 2); "**a sensing layer... a decision layer... and an actuation layer**" (Section 2.1, p. 3); and repeated multi-part sequences such as "**image position, optical path, actuator commands and environmental context**". Again, this is not inherently bad writing; the issue is that it appears so consistently, and in such finished rhetorical packaging, that it reads more like optimised explanatory prose than like an individual author's relaxed scientific register. [filecite turn0file0](#)

On the specific "LLM tells" you asked me to check: **hedge-stacking is not actually the main problem here**. The manuscript does not heavily rely on "it is worth noting that", "it should be emphasized that", "notably", "moreover", or "furthermore". In fact, the prose is often **more assertive than hedge-heavy**. The closer tell is instead **polished contrastive framing, repeated self-scoping, and a consistent explanatory cadence**. The sentence rhythm is not globally uniform, but in the abstract, introduction, discussion, and conclusions there are many long multi-clause sentences with very similar movement: claim → contrast → qualification → restatement. That local regularity is what is most likely to trip suspicion. [filecite turn0file0](#) [filecite turn0file1](#)

My plain-language estimate is this: **a detector is unlikely to label the whole manuscript as cleanly human-authored with high confidence, and it is quite plausible that it would flag portions of the abstract, introduction, discussion, and conclusion as AI-assisted or AI-polished**. I would call the whole-paper risk **moderate**, and the risk for the highest-exposure prose blocks **moderate-to-high**. The **highest-risk passages** are the abstract (especially the second half), the methodological framing in the Introduction, the Discussion paragraph beginning "The value of the architecture reported here...", the paragraph beginning "The open-loop result is the more load-bearing...", the paragraph beginning "The scientific value of this work...", the "An extended architecture under validation" subsection, the "In preparation" future-work section, and the Conclusions, which restage much of the abstract in the same polished contrastive idiom. [filecite turn0file0](#)

## Humanization recommendations

The part that most needs a human pass is the **abstract**, because that is where editorial suspicion starts. Three sentences are especially high-risk. First: "**Disabling the fine-optics stage... indicating that fine-scale stabilisation is carried disproportionately by the fast actuator rather than the slow one.**" Second: "**A genuine open-loop baseline... the first directly measured, rather than inferred, quantification...**" Third: "**The control law, its identified environmental model and its complete safety logic are reported in full, giving a reproducible, guarded architecture whose actuator roles are now supported directly... rather than by inference.**" What makes these feel artificial is not the science but the **stacked contrast clauses, advertising adjectives** ("genuine", "complete", "reproducible", "guarded"), and **cumulative cadence of significance claims**. A human editing pass should reduce the number of contrastive pivots per sentence, strip out adjectives that sell the

architecture rather than describe it, and separate raw result statements from interpretive framing. Do not make it blander; make it less over-compressed and less self-aware. [filecite turn0file0](#)

The **Introduction** would benefit from the next-pass most. The paragraph beginning “**The relevant instrumentation literature can be read as a sequence of methodological lessons rather than as performance benchmarks...**” reads like literature-review prose that has been rhetorically optimised. Likewise, “**The research question is therefore deliberately bounded... The paper addresses this through three evidence-led questions...**” sounds like grant-style or LLM-polished framing rather than normal instrumentation-paper voice. The problem is the **meta-discourse**: the manuscript keeps talking about how carefully it is scoping itself, how it should be read, what it is not claiming, and how the question is framed. A human edit should keep the discipline and caveats, but reduce the amount of explicit narration about those caveats. In this subfield, authors usually earn trust by showing the limitation once and then moving on, rather than restating the epistemic boundaries in polished prose every few paragraphs. [filecite turn0file0](#)

The **Figure 1 caption** is one of the passages most likely to read as AI-enhanced because it switches from figure description to grand interpretation in a very generic register: “**These observations demonstrate that environmental perturbations—particularly pressure and humidity fluctuations—are the dominant source of optical path instability in precision spectroscopic systems. The direct, real-time coupling observed here provides empirical motivation for implementing closed-loop feedback control...**” The mechanism of artificiality here is **generic concluding flourish**. It sounds like an explanatory caption written to “sell the point” rather than a restrained caption written by an instrumentation author. A human pass should keep the descriptive content of the panels and remove or sharply narrow the generalising rhetoric, especially because the surrounding prose on p. 2 actually attributes the visible feature to a **temperature** step while saying pressure and humidity “cannot account for the feature.” [filecite turn0file0](#)

The **Discussion** is the single biggest reservoir of AI-polish signals. Passages such as “**The open-loop result is the more load-bearing of the two new comparisons**”, “**This is a useful negative result rather than a null one**”, and “**The scientific value of this work to the wider instrumentation community is that it demonstrates...**” are not scientifically wrong, but they read like **rhetorical packaging**, not plain expert interpretation. “Load-bearing” is metaphorical and colloquial in a way that sounds imported from a modern internet/LLM register; “useful negative result rather than a null one” is tidy and balanced in a way editors often associate with AI smoothing; and “The scientific value of this work...” is openly promotional. A human editing pass should replace these with more matter-of-fact evidence discussion, cut the self-conscious evaluative phrasing, and let the result stand on the numbers and the experimental design. [filecite turn0file0](#)

The **Section 5.1 / Section 6 material on the extended architecture and planned validation** also needs attention, not because it sounds machine-written in isolation, but because it has the tone of a proposal or design memo rather than a finished journal article. “**An extended architecture, currently undergoing a multi-day validation run...**” followed by “**both are reported here as architecture, not as results**” and then an entire section beginning “**In preparation.**” makes the manuscript sound as though it is trying to narrate what it *will become*, not just what it currently demonstrates. That forward-looking managerial prose is another common LLM polish signal. A human pass should decide what belongs in this paper at all, then keep only the minimum necessary prospective material, in a leaner and less self-explanatory form. [filecite turn0file0](#)

## Scientific professionalism and journal tone

Relative to how instrumentation papers in **RASTI, PASP, AJ, and A&A** usually read, the manuscript is **technically credible but rhetorically over-narrated**. The problem is not lack of professionalism; it is that the prose persistently explains the manuscript's own logic, boundaries, and scientific virtues. In this literature, the strongest papers usually present apparatus, calibration, control law, and performance in a more stripped-down register. Here, the technical core is sounder than the prose style: the equations, command counts, run durations, and explicit limitations are all good signs, but the framing repeatedly steps beyond normal claim-scoping into self-commentary. That is where the tone departs from journal norms. [filecite turn0file0](#)

Several claims are **well motivated but slightly over-scoped relative to the evidence in the same paragraph**. The clearest example is the tension between the p. 2 introductory discussion and the Figure 1 caption. The running text says that “**a small step in the recorded temperature coincides with a distinct dip in the simultaneously monitored optical path**” and that humidity and pressure “**cannot account for the feature**.” The caption, however, concludes that “**pressure and humidity fluctuations —are the dominant source of optical path instability in precision spectroscopic systems**.” Those two statements do not sit comfortably together. The specific two-hour window described in the text supports a local temperature-linked feature; it does not justify a broad caption-level claim about the dominant source in “precision spectroscopic systems” generally. [filecite turn0file0](#)

A second over-scope occurs in Section 4.3, where the manuscript concludes: “**This is a useful negative result rather than a null one... and points back toward actuator saturation, rather than optical-path-model fit, as the more likely source of residual instability in earlier configurations**.” The first half is rhetorical; the second half is an inference that may be plausible, but the paragraph itself does not present a direct actuator-saturation analysis. It would be more professionally toned if it were described explicitly as a tentative interpretation rather than an evidentially preferred explanation. [filecite turn0file0](#)

A third tone issue is **promotional or semi-promotional phrasing**. I would flag the following as out of register for a measured instrumentation paper: “**the more load-bearing of the two new comparisons**” (Discussion, p. 7); “**The scientific value of this work to the wider instrumentation community is that it demonstrates...**” (Discussion, p. 9); “**actually earning its complexity**” (Discussion, p. 9); and “**the first direct, rather than inferred, measurement... in this programme**” in both abstract and conclusions unless the prior programme history is documented robustly enough inside this paper to carry that burden. These phrases sound more like advocacy than normal results prose. [filecite turn0file0](#)

There is also some **inconsistent scoping** around what is a result versus what is under validation. Section 5.1 explicitly introduces an “extended architecture” that is “currently undergoing a multi-day validation run,” then says that the mechanisms are “reported here as architecture, not as results,” and Section 6 begins “In preparation.” That is not inherently improper, but it weakens submission tone because it makes the paper feel partly retrospective and partly aspirational. Reviewers in instrumentation journals typically tolerate future-work paragraphs; they are less enthusiastic about giving a full section of article real estate to work that is still in progress. [filecite turn0file0](#)

## Jargon, adjective, and adverb density

This manuscript's jargon density is **not excessive in the equations/methods sense**; most of the control and instrumentation terminology is appropriate. The more important issue is **rhetorical jargon**:

words that create a polished conceptual wrapper around straightforward technical content. The clearest examples are “**guarded**”, “**self-identifying**”, “**self-validating**”, “**load-bearing**”, “**evidence-led**”, “**reduced-authority mode**”, “**fully passive mode**”, and “**resolved-with-a-caveat**”. Some of these are understandable internally; taken together, they make the manuscript sound as though it is branding its own architecture. In a human revision pass, I would ask the authors to keep only the labels they genuinely need operationally and remove the rest where they are functioning mainly as tone markers.

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The manuscript also has noticeable **adjective/adverb stacking**. Examples include “**reproducible, guarded architecture**” (Abstract, p. 1); “**guarded, event-driven controller**” (Section 3, p. 4); “**guarded, self-identifying, dual-bandwidth controller**” (Discussion, p. 9; Conclusions, p. 10); “**genuine open-loop baseline**” (Abstract, p. 1); “**genuine zero-actuation condition**” (Results/Discussion, pp. 7–8); “**deliberately bounded**” (Introduction, p. 2); “**the more consequential result**” (Results, p. 6); and “**a useful negative result rather than a null one**” (Section 4.3, p. 7). Each extra adjective adds interpretive force, but many add no new measurement content. Their cumulative effect is one of over-insistence.

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There are several places where the prose uses a **vague qualifier where a specific number, threshold, or operational rule appears to exist**. Examples: “**a small ridge parameter  $\lambda$** ” (Section 3.1, p. 4) should state  $\lambda$  if fixed; “**exceeds a floor in magnitude**” and “**once the model quality... is adequate**” suggest numerical gates that are not given on the page (Section 3.1–3.2, pp. 4–5); “**within a few degrees**”, “**within a few per cent**”, and “**near zero**” occur in the fast-channel calibration discussion (Section 3.4, p. 6) and could often be replaced by the actual measured values; “**a larger outdoor-temperature swing**” and “**fresh environmental conditions**” in future-work framing are also vague. When hard numbers exist, replacing these soft qualifiers with the actual limits will not just improve precision; it will also make the prose feel less templated.

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One further diction issue is that some modifiers introduce unnecessary rhetorical weight: “**directly**”, “**genuine**”, “**actually**”, “**deliberately**”, and “**scientific value**” are doing argumentative work more than descriptive work. This is particularly visible in the abstract, discussion, and conclusions. A human editing pass should ask of each such modifier: does it change the scientific meaning, or only turn up the emphasis? If the latter, it is a good candidate for deletion.

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## Structural and cross-reference integrity

There is a **clear broken figure reference**. In the rendered PDF on p. 2 the sentence reads “**Figure ?? shows this coupling directly...**”, which is exactly the kind of visible drafting error that will damage editorial confidence. The source explains why: the text calls `Figure~\ref{fig:thp}`, but the figure is labelled `\label{fig:thp_environmental}`. That must be fixed before submission.

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There is also a **broken citation / undefined reference** in the rendered PDF. Section 3.5 states: “**The spectral method used to characterise the residual optical-path variation in Section 4.2 is the averaged-periodogram method of ?**” on p. 6. In the source, this is `\citet{Welch1967}`. The rendered reference list on pp. 11–12 does not include Welch, so this citation key is evidently unresolved in the compiled manuscript. That is both a technical submission issue and a credibility issue, because it makes the methods section look unfinished.

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A softer structural concern is the manuscript’s inclusion of a full subsection on an “**extended architecture under validation**” and a separate “**Planned validation experiments**” section beginning

**“In preparation.”** Those sections are coherent, but they change the article’s centre of gravity from reporting demonstrated performance to narrating the next phase of development. That can make the manuscript feel less settled and more proposal-like. If the journal allows such forward-looking material, it should still be kept subordinate to the demonstrated experiment. At the moment, the prospective material is prominent enough to feel structurally intrusive. [filecite turn0file0](#)

The abstract and conclusions are broadly supported by the body, but there are two places where support is thinner than the phrasing suggests. First, the repeated **“first direct, rather than inferred”** claim about programme history carries a strong originality burden that is only partly substantiated inside this paper. Secondly, the conclusion that the approach **“transfers to other compact spectrographs”** is plausible as an engineering judgment, but it goes beyond the immediately demonstrated evidence and edges toward generalisation. These are not fatal problems; they are places where the manuscript currently sounds more confident than the in-paper support fully justifies. [filecite turn0file0](#)

## Final verdict

**Yes — as currently written, this manuscript is likely to raise “this reads AI-generated or AI-polished” concerns with at least some editors or reviewers.** My confidence in that verdict is **high**. The risk does **not** come from the science, which reads real and technically engaged. It comes from the prose having been over-smoothed into a recognisable register: too much contrastive framing, too much meta-commentary about scope and contribution, too many repeated branded adjectives for the architecture, and too many paragraph-ending restatements that tell the reader what they have just concluded. The manuscript reads much more like **AI-assisted polishing of real technical work** than like fully AI-generated science. [filecite turn0file0](#)

The **three highest-priority changes**, ranked by impact, are these. **First**, rewrite the **abstract, conclusion, and the most rhetorical discussion paragraphs** back into the authors’ natural technical voice: fewer “rather than” pivots, fewer significance adjectives, fewer self-advertising phrases, and fewer sentences that summarise their own claim. **Second**, strip the **Introduction and Discussion** of excess meta-scoping language — especially the repeated “this paper is not doing X”, “the question is deliberately bounded”, “the scientific value is...”, and similar formulations — while keeping the actual caveats. **Third**, remove or sharply compress the **prospective/under-validation material**, and fix every visible cross-reference/citation failure; unfinished-looking structure amplifies AI suspicion because it makes the polished prose feel even less organically written. [filecite turn0file0](#)  
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If I were acting as a reviewer, my summary would be: **technically interesting and likely publishable after revision, but stylistically over-processed in a way that will draw avoidable AI-authorship suspicion unless the prose is de-smoothed before submission.** [filecite turn0file0](#)

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